

Learning Objectives

At the end of this session, students should be able to:

1. Illustrate the role of modifiable and non-modifiable risk factors in atherosclerotic coronary artery disease and key preventive measures
2. Distinguish cardiac from non-cardiac chest pain
3. Differentiate stable angina from unstable angina
4. Assess the need for PCI versus CABG
5. Demonstrate knowledge of medications used in management of angina pectoris



Defining Chest Pain

2021 AHA/ACC Chest Pain Classification

<u>Category:</u>	<u>Definition:</u>	<u>Clinical Use:</u>
Cardiac Chest Pain (Ischemic)	Pain due to myocardial ischemia (from CAD or coronary vasospasm)	Includes stable angina, unstable angina, NSTEMI/STEMI
Possible Cardiac Chest Pain	Symptoms may represent cardiac ischemia, but are not classic or specific enough	Intermediate-risk cases needing further evaluation
Noncardiac Chest Pain	Clearly not ischemic, or due to other identifiable noncardiac causes	Examples: GERD, costochondritis, panic disorder



Our Patient:

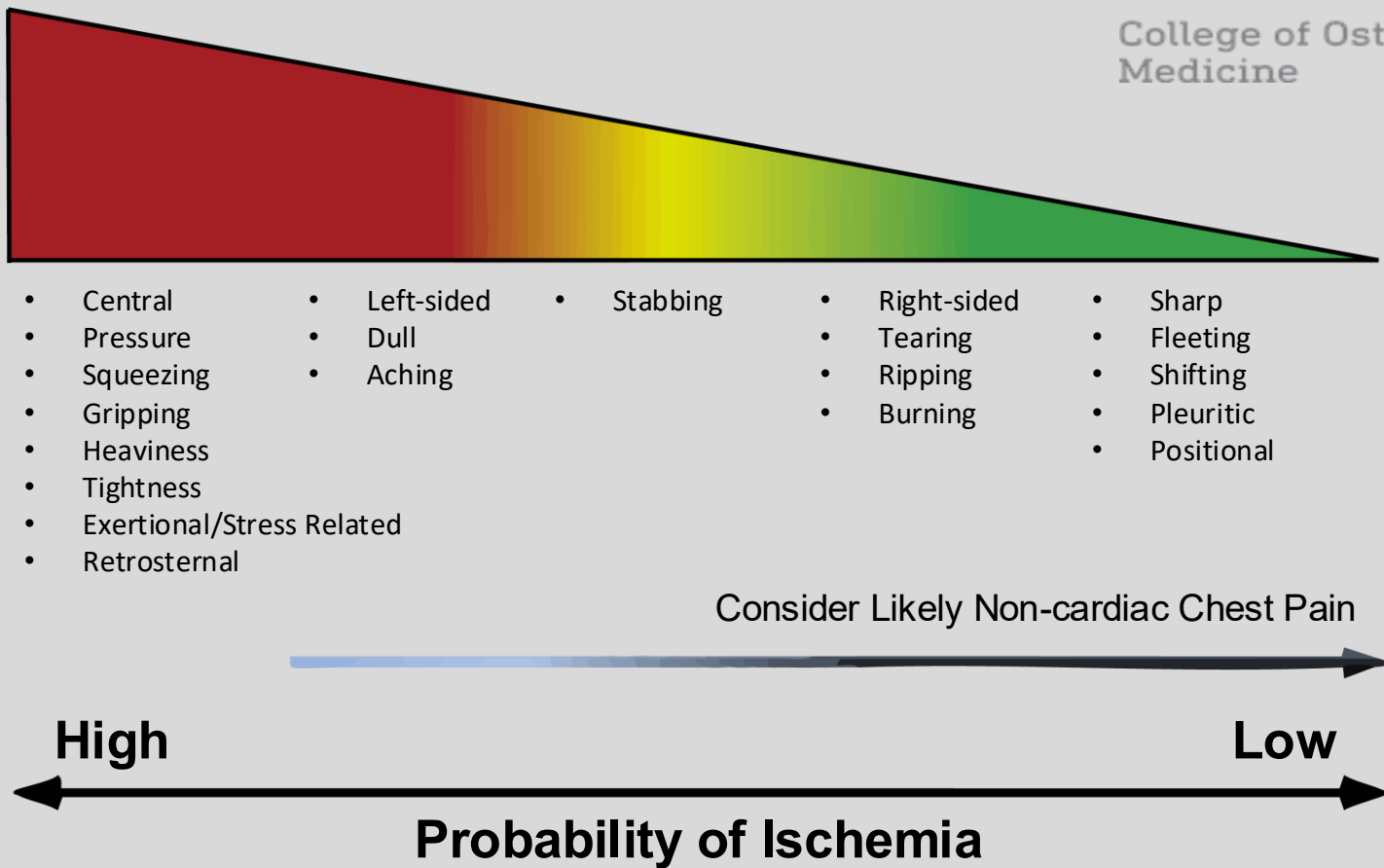
Chief Complaint (CC): “Chest Pain”

History of Present Illness (HPI): A 55-year-old male with a PMHx of HTN and DM type 2 presents to the ED with a 6-month history crushing pain in the middle of his chest. He states that the pain is exacerbated when he goes up the stairs but is relieved with rest. He rates the pain as a 6/10. The pain is present for 7 minutes when he is active. His medications include 20 mg of oral lisinopril daily and 500mg of oral metformin daily. His family history is significant for an MI in his father at the age 50.

What are some differentials for this patient’s presentation?



Is Chest Pain Ischemic In Origin?



Source:
1. 2021AHA/ACC/AHA/ASE/CHEST/SAEM/SCCT/SCMR Guideline for the Evaluation and Diagnosis of Chest Pain J Am Coll Cardiol. 2021 Nov; 78 (22) e187–e285

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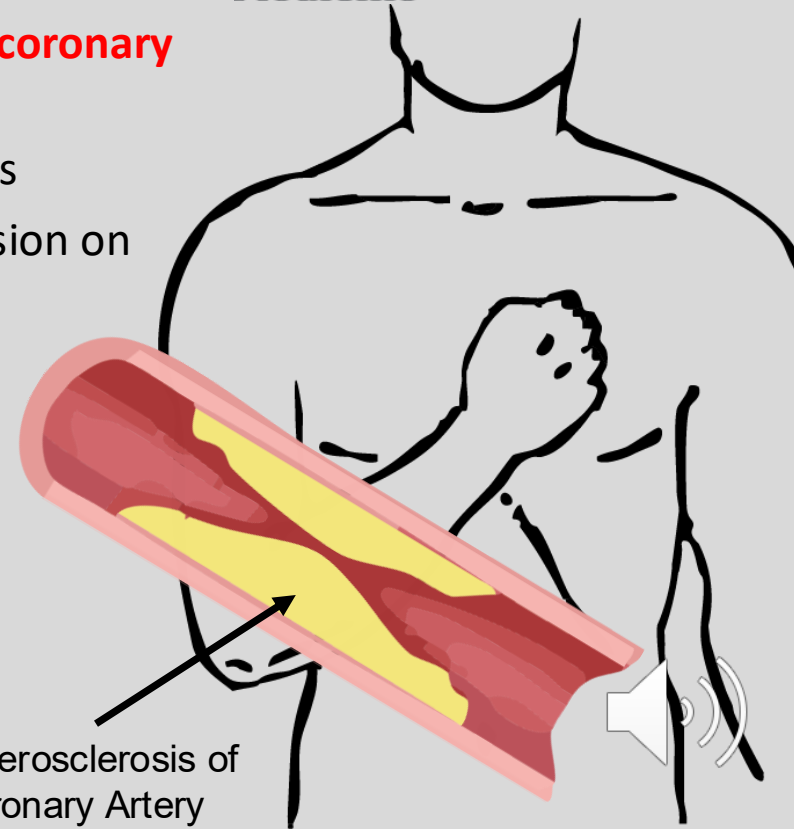


Stable Angina

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OF TECHNOLOGY

College of Osteopathic
Medicine

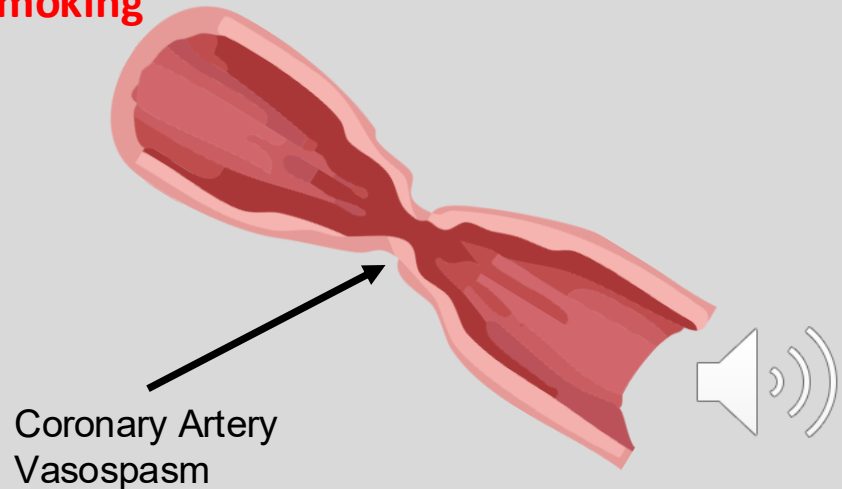
- Usually 2° to atherosclerosis (>70% occlusion) of the coronary artery
- Commonly seen in Men >50 years or woman >60 years
- Episodes of chest discomfort (possibly with ST depression on ECG)
 - Heaviness, Pressure, Squeezing
- Levine's sign: Clenched fist placed over the sternum
- Lasts **2-5 minutes, not more than 10**
- Worse with **Exertion** or emotion (e.g stress)
 - Patient has no symptoms without provocation
- Gets better with **Rest** or **sublingual nitroglycerin**



Atherosclerosis of
Coronary Artery

Vasospastic Angina (Formally Prinzmetal Angina)

- **2° to coronary artery vasospasm**
- **Transient ST-elevation on EKG**
- Associated with migraines and Raynaud's phenomenon
- **Triggers include cocaine, alcohol and triptans**
- **Relieved by nitroglycerin or calcium channel blockers**
- Treatment: **Calcium Channel Blockers, nitrates, smoking cessation (if applicable)**



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What is the workup for this patient?



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Physical Exam: Within normal limits. Labs: LDL of 200mg/dL (Normal: <100mg/dL), HDL of 40mg/dL (Normal: >50mg/dL). The physician decides to perform an exercise stress test.



Question

A 50-year-old female with PMHx of hyperlipidemia presents to the clinic with chest pain for the past 3 months. She describes the pain as “an elephant sitting on my chest.” The pain is brought on by doing chores around the house and relieved with rest. She states the pain does not radiate. The patient states that she smokes ½ pack a day for the past 20 years. Her only medication is 40mg of oral Atorvastatin daily. Vitals and physical exam is normal. There are no changes found on resting ECG. However, on stress testing the patient develops ST segment depression in Leads I and aVL at 85% of her maximal HR. What would be the best appropriate next step in management?

- A. Coronary Artery Bypass Graft (CABG)
- B. Percutaneous Coronary Intervention (PCI)
- C. Start patient on aspirin
- D. Increase the dose of Atorvastatin
- E. Counsel on smoking cessation



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Summary

	<u>Stable Angina</u>	<u>Vasospastic Angina</u>
Pain	Crushing, Squeezing Substernal Pain	Spontaneous crushing chest pain
Exacerbated	Exacerbated with exertion	Unrelated to exertion, can be triggered by cocaine
Relief	Relieved by Rest or Nitroglycerin	Nitroglycerin or CCB
EKG	No EKG findings at rest	Transient ST elevation
Cardiac Biomarkers	None	None
Treatment	PCI, CABG, Lifestyle modification, Nitrates, Beta-blockers/CCB, ACE-Is	CCB are first line

